## 3.2 DIAGNOSTIC TROUBLE CODES

The Teves Mark 20i Antilock Brake System (ABS) module may report any of the following diagnostic trouble codes:

- Controller Failure
- G Switch Not Processable
- Left Front Sensor Circuit Failure
- Left Front Sensor Signal Failure
- Left Rear Sensor Circuit Failure
- Left Rear Sensor Signal Failure
- CAB Power Feed Circuit
- System Over voltage
- System Under voltage
- Pump Motor Not Working Properly
- Right Front Sensor Circuit Failure
- Right Front Sensor Signal Failure
- Right Rear Sensor Circuit Failure
- Right Rear Sensor Signal Failure

Diagnostic trouble codes are retained in memory until erased using the DRBIII®, or automatically erased after 255 key cycles.

### 3.2.1 SYSTEM INITIALIZATION

System initialization starts when the key is turned to RUN. At this point, the CAB performs a complete self-check of all electrical components in the antilock brake systems.

At around 19 km/h (12 miles per hour), the pump motor is briefly activated to ensure operation, and wheel speed sensor circuitry is checked. If the brake pedal is applied when the vehicle reaches 19 km/h (12 mph), this check will be delayed until 40 km/h (25 mph). If, during the dynamic test, the brake pedal is applied, the driver may feel the test through brake pedal pulsations. This is a normal condition. Throughout the remainder of the drive cycle, ongoing tests are performed, and the CAB monitors ABS circuits for continuity.

If any component causes a diagnostic trouble code during system initialization or dynamic check, the CAB will illuminate the ABS warning indicator.

### 3.2.2 DIAGNOSTIC MODE

To enter diagnostic mode, vehicle speed must be below 10 km/h (6 mph), a "no response" message will be displayed by the DRBIII®. The following are characteristics of diagnostic mode:

- The ABS warning indicator will flash. If a hard diagnostic trouble code is present, such as a Speed Sensor Circuit or Signal Failure trouble code or CAB Power Feed Circuit diagnostic trouble code, the indicator will be illuminated without blinking and limited diagnostic operations only will be available until the diagnostic trouble code condition is corrected.
- Antilock operation is disabled.

### 3.2.3 INTERMITTENT DIAGNOSTIC TROUBLE CODES

If the malfunction is not present while performing a test procedure, the diagnostic procedures will not locate the problem. In this case, the code can only suggest an area to inspect. Check for the following:

- loose or corroded connections
- damaged components (sensors, tone wheels especially)
- damaged wiring
- excessive axle shaft runout
- brake system hydraulic leaks
- base brake system problems, non-ABS related.

If no obvious problems are found, erase diagnostic trouble codes and, with the key on, wiggle the wire harness and connectors. Recheck for codes periodically as you work through the system. This procedure may uncover an intermittent or difficult to locate malfunction.

## 3.3 AXLE LOCK

### 3.3.1 GENERAL

For this vehicle, the Axle Lock has no dedicated module controller. A manual switch located on the instrument panel controls the system. No DTC's will occur for this system.

### 3.3.2 DESCRIPTION

The axle lock has the ability to lock the front and rear axle differentials. The system components are the instrument cluster, axle lock switch, front and rear locker relays, front and rear locker pumps, and locker indicator switches. To operate both lockers three requirements need to be met:

1. ignition in the ON position
2. transfer case range sensor indicating 4 "LOW" range
3. vehicle speed less than 16km/h (10 mph)

The 16km/h (10 mph) engagement is for engagement only. All axles will remain locked if vehicle speed does not exceed 72 km/h (45 mph).

The instrument cluster has the control logic that monitors these requirements. If the requirements are not met the axle lock switch will be disabled. The axle lock switch receives the enable signals from the instrument cluster. Any subsequent down-